



Indian Plant Leaf Disease Dataset (Balanced & Augmented)

A comprehensive multi-species plant leaf disease dataset prepared for deep learning research in agriculture, computer vision, and AI-based crop disease detection.



Dataset Overview

Property	Value
Total Images	25,046
Original Images	~7,785
Plant Species	4
Disease Classes	15
Image Resolution	256 × 256 pixels
Image Format	RGB
Dataset Type	Balanced & Augmented



Plant Species Included

SUGARCANE LEAF DISEASE DATASET:

Original Dataset:

- Healthy: 522 images
- Mosaic: 462 images
- RedRot: 518 images
- Rust: 514 images
- Yellow: 505 images
- Total original: 2,521 images

Processed Dataset (sugarcane_processed/):

- Healthy: 1,566 images
- Mosaic: 1,386 images
- RedRot: 1,554 images
- Rust: 1,542 images
- Yellow: 1,515 images
- Total processed: 7,563 images
- Augmentation ratio: 3.00x

Image Properties:

- ✓ Size: 256×256 pixels (with aspect ratio preservation)
- ✓ Channels: 3 (RGB)
- ✓ Pixel range: 0-255
- ✓ Format: Consistent across all images
- ✓ Padding: Applied to maintain square format

❑ BLACKGRAM PLANT LEAF DISEASE DATASET:

Original Dataset:

- Anthracnose 230: 230 images
- Healthy 220: 220 images
- Leaf Crinckle 150: 150 images
- Powdery Mildew 180: 180 images
- Yellow Mosaic 220: 220 images
- Total original: 1,000 images

Processed Dataset (blackgram_processed/):

- Anthracnose 230: 595 images
- Healthy 220: 586 images
- Leaf Crinckle 150: 397 images
- Powdery Mildew 180: 461 images
- Yellow Mosaic 220: 596 images
- Total processed: 2,635 images
- Augmentation ratio: 2.64x

Image Properties:

- ✓ Size: 256×256 pixels (direct resize, no padding needed)
- ✓ Original size: 512×512 (1:1 aspect ratio)
- ✓ Channels: 3 (RGB)
- ✓ Advanced augmentation: Rotation, flip, brightness, zoom, noise, blur

❑ BLACKGRAM PLANT LEAF DISEASE DATASET (ENHANCED):

Original Dataset:

- Anthracnose 230: 230 images
- Healthy 220: 220 images
- Leaf Crinckle 150: 150 images
- Powdery Mildew 180: 180 images
- Yellow Mosaic 220: 220 images
- Total original: 1,000 images

Enhanced Balanced Dataset (blackgrambalanced1000/):

- Anthracnose 230: 1,000 images
- Healthy 220: 1,000 images
- Leaf Crinckle 150: 1,000 images
- Powdery Mildew 180: 1,000 images
- Yellow Mosaic 220: 1,000 images
- Total enhanced: 5,000 images
- Perfect class balance:  1:1:1:1:1 ratio

Image Properties:

- ✓ Size: 256×256 pixels (direct resize from 512×512)
- ✓ Channels: 3 (RGB)
- ✓ Advanced augmentation: 9 techniques applied
- ✓ Multi-augmentation: 1-4 techniques per image
- ✓ Enhancement factor: 5.0x overall



CHILI PLANT DISEASE DATASET:

Original Dataset (ESTIMATED):

- cercospora: ~351 images
- healthy: ~396 images
- mitesandtrips: ~405 images
- nutritional: ~362 images
- powdery mildew: ~362 images
- Total original: ~1,876 images

Processed Dataset (chili_processed/):

- cercospora: 1,055 images
- healthy: 1,188 images
- mitesandtrips: 1,216 images
- nutritional: 1,087 images
- powdery mildew: 1,087 images
- Total processed: 5,633 images
- Augmentation ratio: ~3.0x

Image Properties:

- ✓ Size: 256×256 pixels (with aspect ratio preservation)
- ✓ Channels: 3 (RGB)
- ✓ Well-balanced classes
- ✓ Comprehensive augmentation applied



SOYBEAN PLANT DISEASE DATASET:

Original Dataset (SEVERELY IMBALANCED):

- Disease_Preprocessing data: 1,055 images (88.4%)
- Healthy_Preprocessing data: 139 images (11.6%)

- Total original: 1,194 images

- Imbalance ratio: 7.59:1 ⚠

Balanced Dataset (soybean_balanced/): • Disease_Preprocessing data: 1,055 images (50.0%)

- Healthy_Preprocessing data: 1,055 images (50.0%)

- Total balanced: 2,110 images

- Balance achieved: ✅ 1:1 ratio

Image Properties: ✓ Size: 256×256 pixels (already preprocessed)

✓ Channels: 3 (RGB)

✓ File size: 0.01-0.02 MB (very compressed)

✓ Critical balancing: 916 augmented healthy images created

Preprocessing Pipeline

The dataset was standardized and optimized for deep learning training.

Image Processing

- Resized to 256×256 pixels
- RGB format normalization
- Aspect ratio preservation
- Padding for non-square images

Data Augmentation

Applied augmentation techniques:

- Rotation
- Horizontal Flip
- Brightness Adjustment
- Zoom
- Blur
- Noise Injection
- Hue/Saturation Shift
- Shear Transformations

Dataset Balancing

- Severe class imbalance corrected
- Blackgram dataset enhanced to 1000 images per class

- Soybean dataset balanced from 7.6:1 → 1:1

Dataset Structure

```
dataset/
├── sugarcane_mosaic/
├── sugarcane_yellow/
├── sugarcane_rust/
├── sugarcane_redrot/
├── sugarcane_healthy/
├──
├── blackgram_healthy/
├── blackgram_leafcrinkle/
├── blackgram_powderymildew/
├── blackgram_yellowmosaic/
├── blackgram_anthracnose/
├──
├── chili_powdery_mildew/
├── chili_nutritional/
├── chili_mites_and_trips/
├── chili_healthy/
├── chili_cercospora/
├──
├── soybean_healthy/
└── soybean_diseased/
```

COMPREHENSIVE SUMMARY:

Dataset Statistics:

- Total Original Images: ~7,785
- Total Processed Images: ~25,046
- Overall Augmentation Ratio: ~3.2x
- Number of Plant Species: 4 (Sugarcane, Blackgram, Chili, Soybean)
- Number of Disease Classes: 15 total
- All Images Standardized: 256×256 pixels

Processing Techniques Applied:

- Resize with aspect ratio preservation (Sugarcane, Chili)
- Direct resize for square images (Blackgram, Soybean)
- Padding to maintain square format
- Data augmentation (rotation, flip, brightness, zoom)
- Advanced augmentation (noise, blur, hue/saturation, shear)
- Class balancing for imbalanced datasets
- Enhanced multi-augmentation techniques
- Normalization to [0,1] pixel values

Critical Issues Resolved:

- Sugarcane: Large dataset standardized and augmented
- Blackgram: Enhanced to perfect 1000 images per class
- Chili: Well-balanced augmentation across all disease types
- Soybean: CRITICAL class imbalance fixed (7.6:1 → 1:1)

Dataset Breakdown by Processing Quality:

- PREMIUM: blackgrambalanced1000/ (5,000 images, perfect balance)
- EXCELLENT: chili_processed/ (5,633 images, well balanced)
- EXCELLENT: sugarcane_processed/ (7,563 images, comprehensive)
- GOOD: soybean_balanced/ (2,110 images, critical fix applied)

INITIAL DATA SOURCES:

The following datasets were sourced from public repositories:

Sugarcane Leaf Disease Dataset:

- Source: <https://www.kaggle.com/datasets/nirmalsankalana/sugarcane-leaf-disease-dataset>
- Contains: Healthy, Mosaic, RedRot, Rust, Yellow diseases

☐ Blackgram Plant Leaf Disease Dataset:

- Source: <https://data.mendeley.com/datasets/zfcv9fmrgv/3>
- Contains: Anthracnose, Healthy, Leaf Crinkle, Powdery Mildew, Yellow Mosaic

Chili Plant Disease Dataset:

- Source: <https://data.mendeley.com/datasets/w2r855hpx8/2>
- Contains: cercospora, healthy, mitesandtrips, nutritional, powdery mildew

Soybean Plant Disease Dataset:

- Source: <https://data.mendeley.com/datasets/tf9dtfz9m6/2>
- Contains: Disease and Healthy classifications

READY FOR CNN TRAINING:

All datasets are now optimized for deep learning: Consistent image dimensions (256×256) Balanced class distributions Sufficient sample sizes (1000+ per class in premium datasets) Realistic augmentations applied Proper preprocessing completed Multiple processing quality tiers available

TRAINING RECOMMENDATIONS:

Model Architecture Suggestions: • **Medium datasets** (soybean): ResNet18/34 with transfer learning • **Large datasets** (others): ResNet50, EfficientNetB0-B3, or Vision Transformer

Class Balance Handling:

- **Perfectly balanced** (blackgram*balanced1000*): Standard training
- **Well balanced** (chili, sugarcane): Standard or light class weights
- **Previously imbalanced** (soybean): Monitor both classes equally

Training Strategy:

- **Single species:** Train specialist models per crop
- **Multi-species:** Train universal plant disease detector
- **Ensemble:** Combine specialist models for best accuracy
- **Progressive:** Start with balanced datasets, add complexity

Performance Monitoring:

- Monitor precision, recall, F1-score for each class
- Use confusion matrices to identify problematic classes
- Validate across all plant species for robustness
- Consider cross-species validation for generalization

Advanced Techniques:

- Apply mixup or cutmix for additional regularization
- Use test-time augmentation for inference
- Implement graduate self-training for unlabeled data
- Consider meta-learning for few-shot disease detection

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